Name
Date
Course

The Exponent Rules (Part II) Homework

Simplify each expression below and write the Roman numerals of the exponent rules that you used in the margin. Do not leave your answers with fraction exponents or negative exponents.

1. $x^5 \cdot x^{-5}$
2. $(5y^4)^3$
3. $\dfrac{y^{10}}{y^9}$
4. $(16)^{-\frac{1}{4}}$
5. $\left(\dfrac{x^5}{y^2}\right)^4$
6. $(y^8)^{\frac{1}{8}}$
7. $\dfrac{4^{-7}}{4^{-7}}$
8. $x^{-8} \cdot x^3$
9. $(8^5)^{\frac{1}{3}}$
10. $\dfrac{a^2}{a^7}$
11. $(x^2y^3)^4$
12. $6^{\frac{1}{2}} \cdot 6^{\frac{1}{8}}$
13. $b^{\frac{1}{3}} \cdot b^{\frac{2}{3}}$
14. $(5^{\frac{2}{7}})^{\frac{3}{4}}$
15. $\left(\dfrac{x^{-6}}{a^{-3}}\right)^{-2}$
16. $4^8 \cdot 4^{-7}$
17. $\dfrac{y^{\frac{5}{6}}}{y^{\frac{3}{8}}}$
18. $\left(x^{-\frac{7}{5}}\right)^{-\frac{5}{7}}$
19. $64^{-\frac{9}{2}} \cdot 64^6$
20. $(6y)^{-2}$
21. $\dfrac{y^{-\frac{1}{3}}}{y^{-\frac{4}{3}}}$
22. $(4^2)^{-1}$
23. $(b^{-5})^4$
24. $x^{\frac{3}{5}} \cdot x^{\frac{2}{5}}$
25. $4^{-\frac{1}{6}} \cdot 4^{\frac{1}{6}}$
26. $(ab^3c^8)^2$
27. $(8^{-3} \cdot 8^7)^{\frac{1}{3}}$
28. $(81^{-2})^{\frac{1}{4}}$
29. $(x^3 \cdot x^4 \cdot b^2)^2$
30. $(a^2 \cdot a)^3$
31. $\left(\dfrac{xc^6}{c^2}\right)^3$
32. $(x^3 \cdot y^5 \cdot x^{-3})^3$
33. $(b^{-8} \cdot b^9)^{\frac{1}{2}}$

(34) $\left(\dfrac{x^8}{y^8}\right)^{\frac{1}{8}}$

(35) $\left(\dfrac{ab^{10}}{b^{10}}\right)^{\frac{3}{4}}$

(36) $\left(\dfrac{a^8b^4}{a^3b^2}\right)^5$

(37) $(2x^6)^{-4}$

(38) $\left(\dfrac{y^7}{y^6}\right)^{-3}$

(39) $\sqrt{25^2\cdot 25}$

(40) $\sqrt[4]{(16^{\frac{1}{5}})^{10}}$

(41) $\sqrt[3]{\dfrac{125^6}{125^4}}$

(42) $\sqrt[3]{8\cdot 8^4}$

Answers

I, VII (1) $\boxed{1}$

III, V (2) $\boxed{125y^{12}}$

II, VI (3) $\boxed{y}$

VIII, IX (4) $\boxed{\dfrac{1}{2}}$

IV, V (5) $\boxed{\dfrac{x^{20}}{y^8}}$

V, VI (6) $\boxed{y}$

II, VII (7) $\boxed{1}$

I, VIII (8) $\boxed{\dfrac{1}{x^5}}$

V, X (9) $\boxed{32}$

II, VIII (10) $\boxed{\dfrac{1}{a^5}}$

III, V (11) $\boxed{x^8y^{12}}$

I, X (12) $\boxed{(\sqrt[8]{6})^5}$

I, VI (13) $\boxed{b}$

V, X (14) $\boxed{(\sqrt[14]{5})^3}$

IV, V (15) $\boxed{\dfrac{x^{12}}{a^6}}$

I, VI (16) $\boxed{4}$

II, X (17) $\boxed{(\sqrt[24]{x})^{11}}$

V, VI (18) $\boxed{x}$

I, X (19) $\boxed{512}$

VIII, III (20) $\boxed{\dfrac{1}{36y^2}}$

II, VI (21) $\boxed{y}$

V, VIII (22) $\boxed{\dfrac{1}{16}}$

V, VIII (23) $\boxed{\dfrac{1}{b^{20}}}$

I, VI (24) $\boxed{x}$

I, VII (25) $\boxed{1}$

III, V (26) $\boxed{a^2b^6c^{16}}$

I, V, X (27) $\boxed{16}$

V, VIII, IX (28) $\boxed{\dfrac{1}{9}}$

I, III, V (29) $\boxed{x^{14}b^4}$

VI, I, V (30) $\boxed{a^9}$

II, III, V (31) $\boxed{x^3c^{12}}$

I, VII, V (32) $\boxed{y^{15}}$

I, V, IX (33) $\boxed{\sqrt{b}}$

IV, V, VI (34) $\boxed{\dfrac{x}{y}}$

II, VII, X (35) $\boxed{\left(\sqrt[4]{a}\right)^3}$

II, III, V (36) $\boxed{a^{25}b^{10}}$

VIII, III, V (37) $\boxed{\dfrac{1}{16x^{24}}}$

II, V, VIII (38) $\boxed{\dfrac{1}{y^3}}$

VI, I, X' (39) $\boxed{125}$

V, X' (40) $\boxed{4}$

II, X' (41) $\boxed{25}$

VI, I, X' (42) $\boxed{32}$